

Introduction to Big Data

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What is Big Data?

Big Data refers to complex and large data sets that have to be processed and analyzed to uncover valuable information that can benefit businesses and organizations.

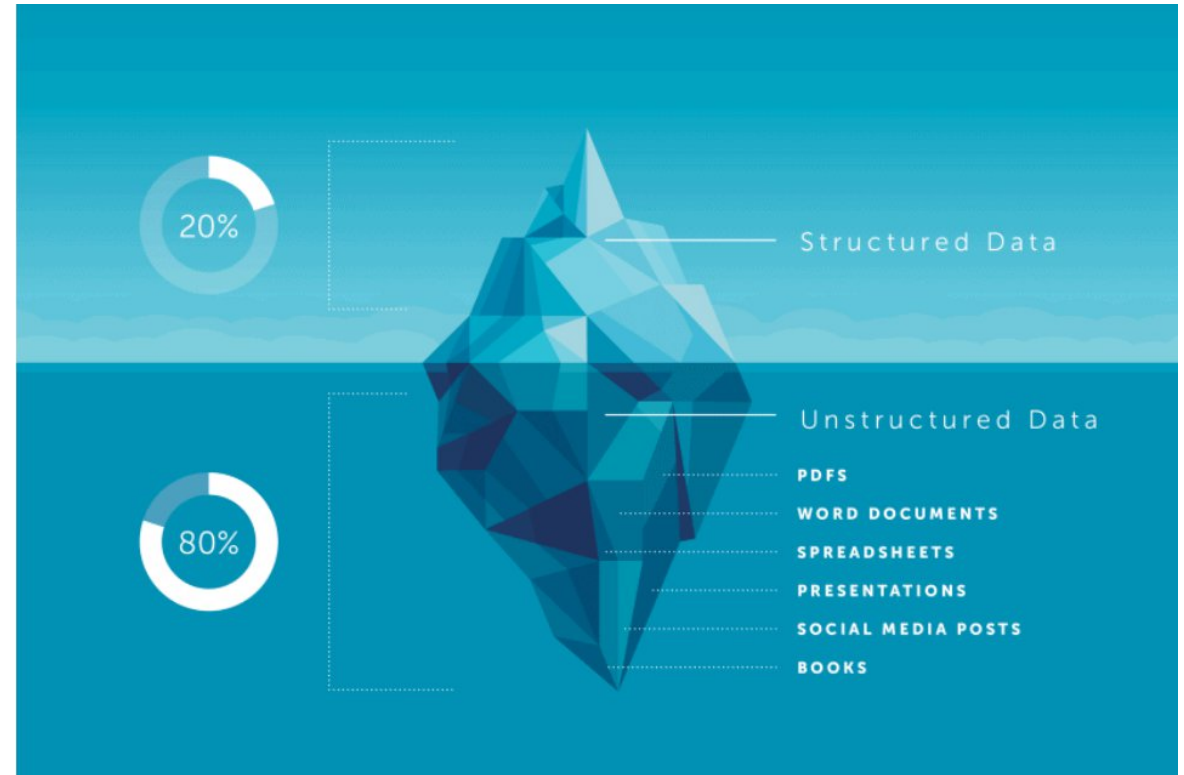
It encompasses the volume of information, the velocity or speed at which it is created and collected, and the variety or scope of the data points being covered (known as the "three v's" of big data).

What is Big Data?

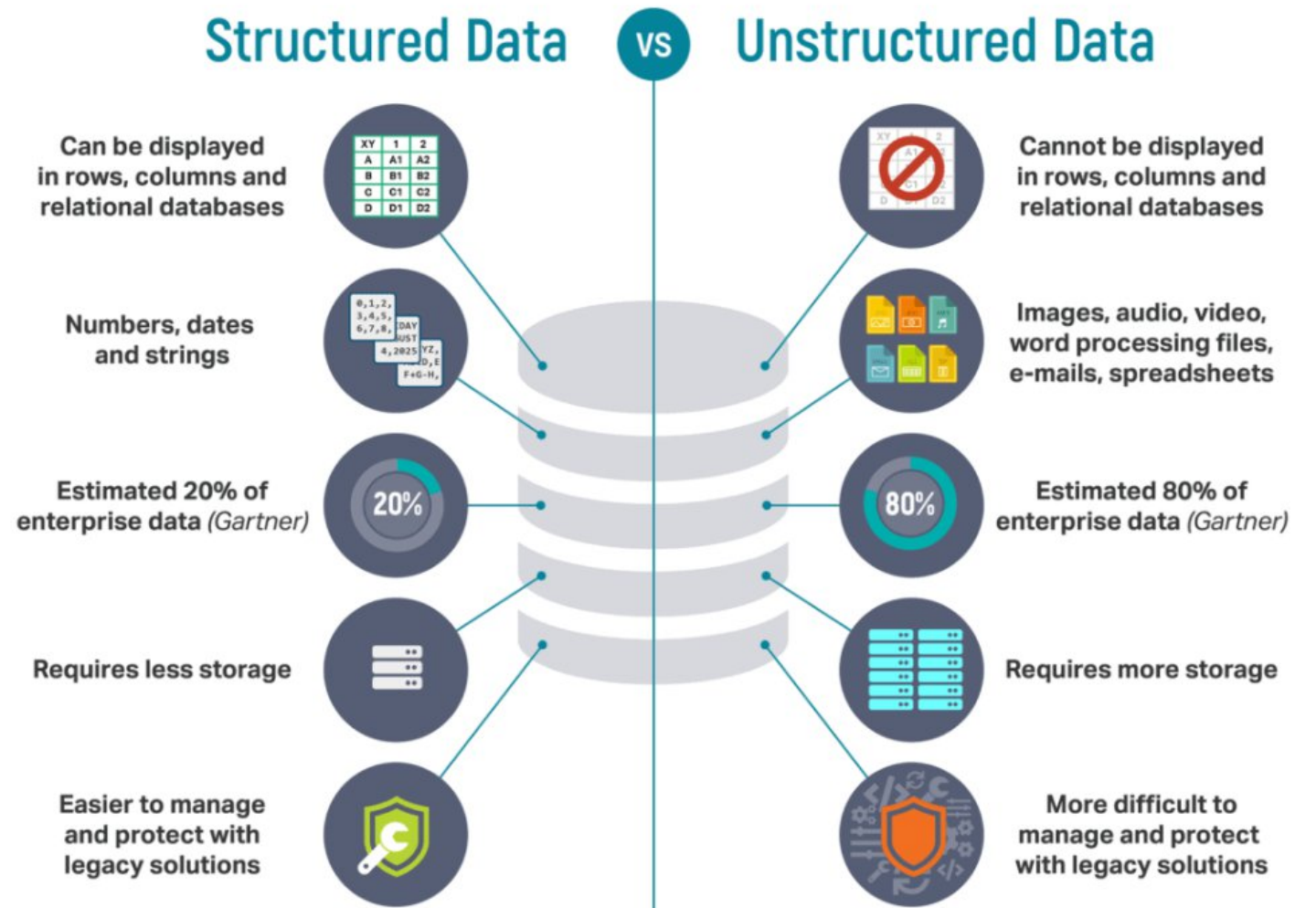
- The increase in the amount of data available presents both opportunities and problems.
- In general, having more data on customers (and potential customers) should allow companies to better tailor products and marketing efforts in order to create the highest level of satisfaction and repeat business.
- Companies that collect a large amount of data are provided with the opportunity to conduct deeper and richer analysis for the benefit of all stakeholders.



Structured Data and Unstructured Data



Structured Data and Unstructured Data



Structured Data

Structured data is generally tabular data that is represented by columns and rows in a database.

Databases that hold tables in this form are called relational databases.

The mathematical term “relation” specifies a formed set of data held as a table.

In structured data, all row in a table has the same set of columns.

SQL (Structured Query Language) programming language used for structured data.

Structured Data

id	name	age
1	Jim	28
2	Pam	26
3	Michael	42

id	subject	Teacher
1	Languages	John Jones
2	Track	Wally West
3	Swimming	Arthur Curry
4	Computers	Victor Stone

student_id	subject_id	grade
2	1	98
1	2	100
1	4	75
3	3	60
2	4	76
3	2	88

Unstructured Data

Unstructured data is information that either is not organized in a pre-defined manner or does not have a pre-defined data model.

Unstructured information is a set of text-heavy but may contain data such as numbers, dates, and facts as well.

Videos, audio, and binary data files might not have a specific structure. They're assigned to unstructured data.

Unstructured Data



Text files and documents



Server, website, and applications logs



Sensor data



Image files



Video files



Audio files



Emails



Social media data

Semi Structured Data

Semi-structured data is information that doesn't consist of Structured data (relational database) but still has some structure to it.

Semi-structured data consists of documents held in JavaScript Object Notation (JSON) format and XML etc.

It also includes key-value stores and graph databases.

Semi Structured Data

```
## Document 1 ##
{
  "customerID": "103248",
  "name":
  {
    "first": "AAA",
    "last": "BBB"
  },
  "address":
  {
    "street": "Main Street",
    "number": "101",
    "city": "Acity",
    "state": "NY"
  },
  "ccOnFile": "yes",
  "firstOrder": "02/28/2003"
}
```

```
<studentsList>
  <student id="1">
    <firstName>Greg</firstName>
    <lastName>Dean</lastName>
    <certificate>True</certificate>
    <scores>
      <module1>70</module1>
      <module12>80</module12>
      <module3>90</module3>
    </scores>
  </student>
  <student ind="2">
    <firstName>Wirt</firstName>
    <lastName>Wood</lastName>
    <certificate>True</certificate>
    <scores>
      <module1>80</module1>
      <module12>80.2</module12>
      <module3>80</module3>
    </scores>
  </student>
</studentsList>
```

Sources of Big Data

A significant part of big data is generated from three primary resources:

- Machine data
- Social data, and
- Transactional data

In addition to this, companies also generate data internally through direct customer engagement.

This data is usually stored in the company's firewall.

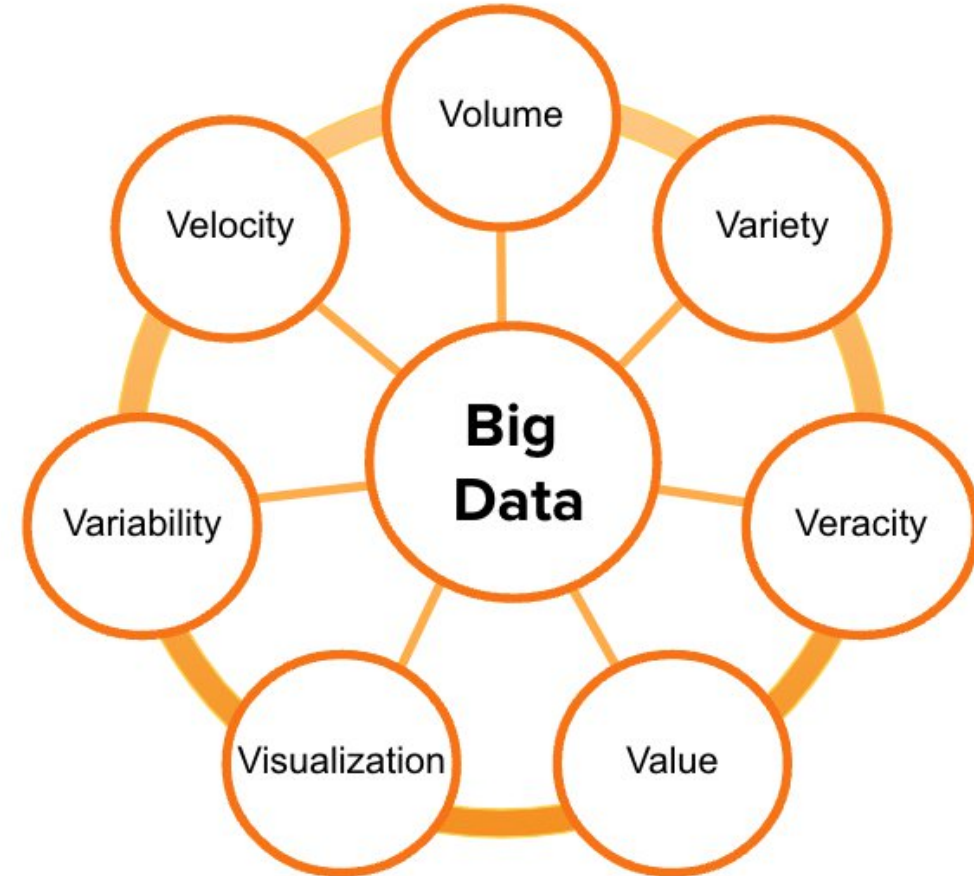
It is then imported externally into the management and analytics system.

Dimensions of Big Data – 7V s

- Big Data systems have “data sets characterized by their volume, velocity of change and variety of type (the ‘Three Vs’)”

(Economist Intelligence Unit 2013)

7 V'S OF BIG DATA



Volume : Scale of Data

- This refers to the vast amounts of data generated every second.
- This is not just Terabytes (10^{12} or 10^6) but Zettabytes (10^{21}) or Yottabytes (10^{24}), or a new Big data word: Brontobytes (1,000 Yottabytes).
- If we take all the data generated in the world between the beginning of time and 2008, the same amount of data will soon be generated every minute.
- New big data tools use distributed systems so that we can store and analyze data across databases that are dotted around anywhere in the world.



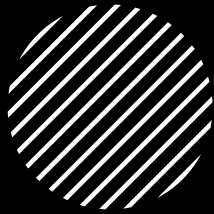


Velocity : Analysis of Streaming Data

- This refers to the speed at which new data is generated and the speed at which data moves around.
- Viral advertising is now a method of advertising which relies on digital transmission
- Viral ads generally take the form of some impressive video footage or something 'chat-worthy' which is designed to be passed through the population via email or a similar tool.
- Many people tend to send viral ads to a large group of friends which is why viral can spread quickly and effectively across the world
- Technology allows us now to analyze such data while it is being generated without ever putting it into databases



Variety : Different Forms of Data



- This refers to the different types of data we can now use.
- In the past, we focused on structured data that fit neatly into tables or relational databases, such as financial data.
- In fact, 80% of the world's data is unstructured (text, images, video, voice, etc.)
- Big data technology means we can now analyze and bring together data of different types such as messages, social media conversations, photos, sensor data, and video or voice recordings.



Veracity : Uncertainty of Data

- Refers to the messiness or trustworthiness of the data.
- With many forms of big data quality and accuracy are less controllable
 - Think of Twitter posts with hash tags, abbreviations, typos, and colloquial speech as well as the reliability and accuracy of the content
- Technology now allows us to work with this type of data.

Turning Big Data into Value

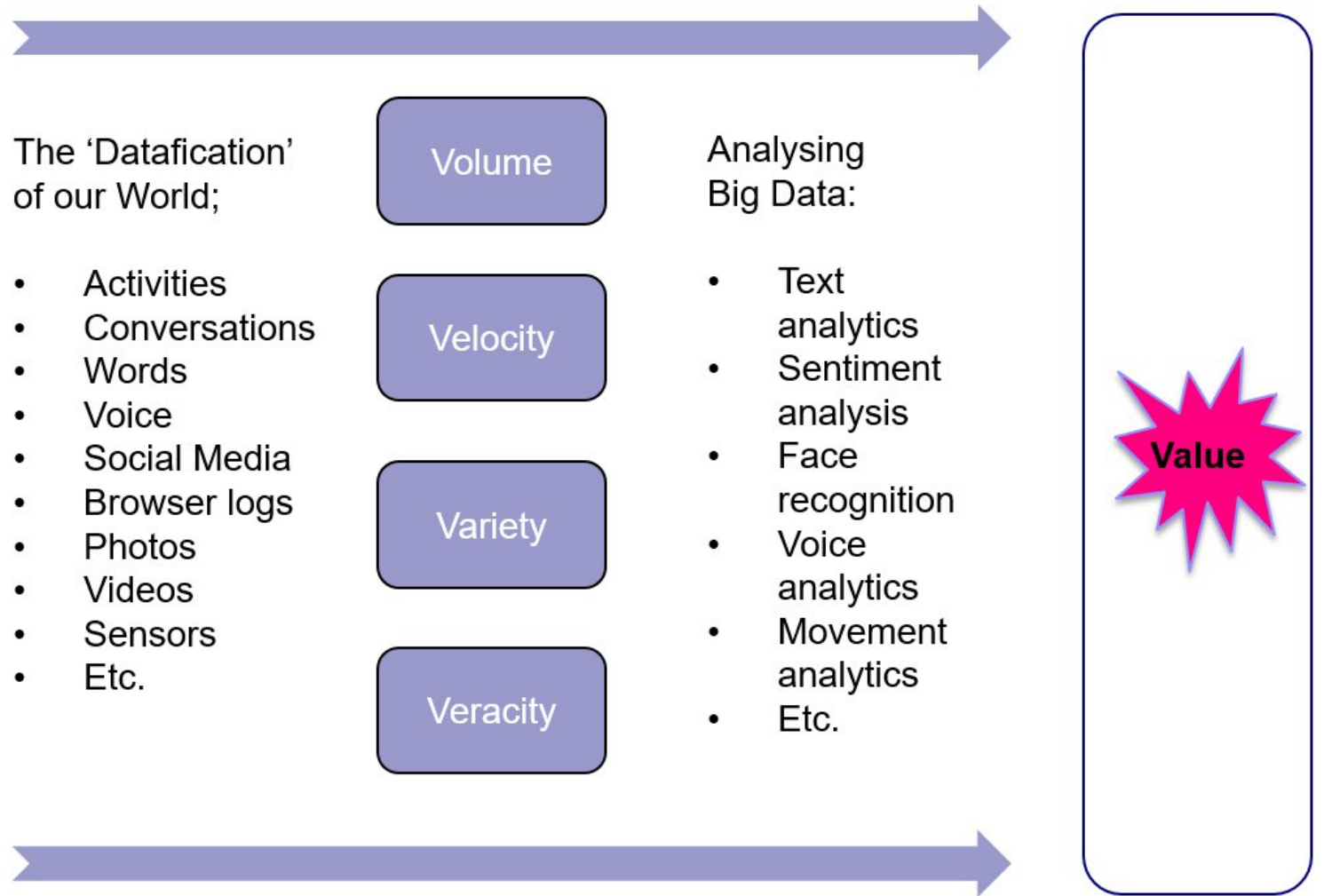
The “Datafication” of our world gives us unprecedented amounts of data in terms of Volume, Velocity, Variety, and Veracity.

The latest technology helps gain insights and add value.

Such technologies:

- Cloud Computing
- Distributed Systems
- Grid Databases
- Software And Analysis Approaches

Turning Big Data into Value



More V's

Variability

- This refers to the complexity of the data set
- In comparison with Variety, it means the number of variables in the data sets.

Visibility/ Visualization

- This empathizes that you need to have a full picture of the data in order to make informative decisions